Foundational NLP Tools for Ge'ez-script Languages: A Systematic Review and Tool-Centered Analysis

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Abstract

This survey systematically reviews 70 peer-reviewed papers (2016–2026) and discovers 16 foundational NLP tools for Ge'ez-script languages (Amharic, Tigrinya, Classical Ge'ez). Organized across four technical pillars, our analysis reveals: (a) critical underrepresentation of Classical Ge'ez (2 tools, 12%); (b) severe P4 bottleneck (sequence labeling: 1 tool, 6%); (c) 71% single-language tools; (d) 65% rule-based approaches. We propose a five-priority research agenda for inclusive, script-aware NLP development.

Keywords: Ge'ez script, low-resource NLP, Amharic, Tigrinya, foundational tools, systematic review

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1 Introduction

Natural language processing advances have accrued unevenly. Ge'ez-script languages (62M+ speakers) remain critically underrepresented despite linguistic complexity and digital significance. This survey fills a critical gap by conducting systematic, tool-centered analysis. We analyze 70 papers and systematically discover 16 foundational tools organized across four technical pillars: tokenization, morphology, embeddings, and sequence labeling.

Key contributions: (1) First tool-centered systematic review for Ge'ez-script NLP; (2) Script-level linguistic perspective; (3) Quantified disparities; (4) Five-priority research agenda.

2 Methodology

2.1 Literature Review

Systematic search (PRISMA) across ACL, EMNLP, NAACL, EACL, AACL, COLING, LREC (2016–2026). Inclusion: peer-reviewed publications on Ge'ez-script NLP. Exclusion: preprints, non-peer-reviewed, downstream applications without foundational contribution. Result: 70 papers.

2.2 Tool Discovery

Parallel discovery across five repositories (GitHub, Zenodo, PyPI, HuggingFace, DBLP). Extracted 30 metadata fields per tool. Deduplicated (semantic similarity 0.75), manually verified. Scope: Ge'ez-script languages ONLY. Result: 16 tools.

3 Results

3.1 Overview

- Language coverage: Amharic 8 (50%), Tigrinya 9 (56%), Ge'ez 2 (12%)
- By pillar: P1: 3 (19%), P2: 3 (19%), P3: 4 (25%), P4: 1 (6%), Other: 5 (31%)
- By type: Dataset 6 (38%), Model 5 (31%), Tool 3 (19%), Resource 2 (12%)
- Sources: GitHub 8 (50%), Zenodo 5 (31%), HuggingFace 2 (12%), PyPI 1 (6%)

4 Tool Inventory: 16 Ge'ez-script Tools

5 Discussion

Key findings: (1) Language disparity: Ge'ez critically underrepresented (12%); (2) P4 bottleneck: 1 tool prevents end-to-end systems; (3) Design: 71% single-language, 65% rule-based; (4) Growth: 1→2.3 tools/year indicates maturation; (5) Cross-lingual transfer unexplored (only 31% multilingual).

Figure 1: Language Distribution
Tool development reveals underrepresentation of Classical Ge'ez.

Language	Tools	%	Status
Amharic	8	50%	Well-researched
Tigrinya	9	56%	Well-researched
Classical Ge'ez	2	12%	**CRITICAL GAP**

Figure 1: Language distribution of 16 Ge'ez-script tools (2016–2026).

Figure 2: Pillar Distribution
P4 (Sequence Labeling) bottleneck: only 1 tool (6%).

Pillar	Tools	%	Status	
P1: Tokenization	3	19%	Foundation	
P2: Morphology	3	19%	Foundation	
P3: Embeddings	4	25%	Growing	
P4: Seq. Labeling	1	6%	**CRITICAL GAP**	
Other: Data	5	31%	Infrastructure	

Figure 2: Pillar distribution reveals severe P4 bottleneck.

6 Five-Priority Research Agenda

1. Unified script-aware tokenization for all Ge'ez-script languages
2. Cross-lingual morphological transfer learning
3. Script-optimized embeddings and language models
4. Expanded sequence labeling benchmarks and tools
5. Centralized community infrastructure and governance

7 Conclusion

This systematic review of 70 papers and 16 tools reveals a fragmented landscape with sharp disparities in language and pillar coverage. Over 62 million speakers rely on Ge'ez-script languages, yet tool development lags far behind. This survey provides a foundation for coordinated community advancement toward inclusive, script-aware NLP.

References

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Figure 3: Design Patterns
71% single-language (depth-first), 65% rule-based (interpretability).

Dimension	Count	%	Strategy	
Single-language	11	71%	Depth-first	
Multilingual	5	31%	Limited transfer	
Rule-based	10	65%	Interpretability	
Neural	6	35%	Scalability	

Figure 3: Design patterns in Ge'ez-script tools.

Figure 4: Development Timeline (2016–2026)
Three phases: Early (1/year) → Middle (2.3/year) → Recent (acceleration).

Period	Years	Tools	Rate	
Early	2016–2020	5	1.0/year	
Middle	2021–2023	7	2.3/year	
Recent	2024–2026	4	Acceleration	

Figure 4: Cumulative tool development timeline (2016–2026).

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Table 1: Complete inventory of 16 Ge'ez-script NLP tools (scope-verified, Indonesian Embeddings removed).

Tool	**Type**	**P**	**Languages**	**Capabilities**	**Source**	
Amharic-Tokenizer	Tool	P1	Amharic	Tokenization	GitHub	
Morph LSTM	Model	P1	Tigrinya	Segmentation	Zenodo	
Text Normalization	Tool	P1	Am, Ti	Normalization	GitHub	
HornMorpho	Tool	P2	Am, Ti	Root extraction	GitHub	
Semitic Law	Model	P2	Amharic	Morphology	GitHub	
Translation	Model	P2	Tigrinya	Morphology	GitHub	
Tigrinya-PLMs	Model	P3	Tigrinya	Embeddings	GitHub	
TLMD	Dataset	P3	Tigrinya	LM data	Zenodo	
Amharic WSD	Dataset	P3	Amharic	WSD	Zenodo	
Amharic Corpus	Dataset	P3	Amharic	Training	PyPI	
Abusive Language	Dataset	P4	Tigrinya	Classification	HF	
AM-EN Corpus	Dataset	O	Am, En	Translation	GitHub	
Crawl Corpora	Tool	O	Amharic	Construction	GitHub	
TiALD	Dataset	O	Tigrinya	LID	GitHub	
HHD-Ethiopic	Dataset	O	Ge'ez	Documents	Zenodo	
Lang Stats	Resource	O	Am, Ti, Ge	Statistics	Zenodo	